The Open Source Firefighter

A Community Adoption Briefing for Towns, Fire Districts, and Departments

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Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

To the councils, boards, and chiefs reading this

You have seen documents like this one before. A company arrives with a product, a subscription, and a model ordinance, and asks your community to adopt something you will then depend on them for. This document follows that format because the format works — but it is asking for something different, because it is not selling anything.

The Open Source Firefighter is a published open standard for coordinated neighborhood ember defense. There is no company behind it. There is no subscription, no license fee, no vendor to depend on, and no data leaving your community. The specification is free, permanent, and belongs to everyone it protects. What it needs from your community is not a purchase order. It is recognition, a pilot, and a place in your planning.

The problem your community already knows

When wildfire reaches a community, most homes are not lost to the wall of flame. They are lost to embers — burning debris blown ahead of the fire, sometimes by miles — landing on roofs, in gutters, in dry landscaping, starting small fires that become house fires while the front is still distant. In a wind event, thousands of embers fall across a neighborhood at once. No department, anywhere, has enough crews to stand at every house.

The defense is simple and proven: surfaces that stay wet during the ember window do not ignite. Home sprinkler protection has existed for years. What has never existed is coordination — because if every house on a street opens its sprinklers at once, the water system that supplies all of them collapses, and then no one is protected. Defending a neighborhood means homes taking turns on a shared supply, automatically. That is what this system does, and it is the only thing like it.

If your community is in Colorado, you don’t need the theory — you have the Black Forest fire. Nearly five hundred homes, two lives, and a lesson the research has since made explicit: mitigated homes survived at higher rates and it still wasn’t enough, because at neighborhood spacing, homes ignite each other. The community, not the house, is the unit of survival — and until now, no infrastructure existed for a community to defend itself as one. Homeowner planning is supported by a one-page supply and coverage table (conservative figures, ordinary garden-hose arithmetic), and every unit verifies its own actual supply at commissioning.

What it is

A weatherproof box the size of a breaker panel, mounted on a wall. Water in one side, up to three zones of ordinary lawn sprinklers out the other. Built entirely from standard industrial parts — the same controllers, valves, and batteries in your water plant and your fire alarm panels — for roughly $800–1,500 in components, by any competent electrician, irrigation contractor, or capable homeowner.

- **It takes turns.** One zone at a time, rotating, so an ordinary household water connection protects a whole structure. The minimum installation is one garden hose and one hardware-store sprinkler. Two or three protect a mobile home — the housing your most vulnerable residents live in, and the housing proprietary systems ignore.
- **It coordinates the street.** Each box carries a small license-free radio. Boxes on a street schedule around the shared water supply automatically — no internet, no cloud, no account. They talk directly to each other, and they work alone if they have to.
- **It answers to your firefighters.** Crews arriving on scene see every participating home as one picture, on the mapping tools many departments already use, with command authority — including a one-button order for all municipal-supplied units to yield the mains when an engine company needs the water. Homeowners always keep physical control of their own equipment.

- **It keeps records.** Every unit maintains a tamper-evident log. After an event, your community has a verifiable account of what ran, when, and with what water — for your review, your residents' insurers, and your planning.
- **It speaks.** Every unit announces its status in plain language, like a fire panel. The rugged field version talks a first-time user through its own setup — a mutual-aid volunteer can put one in service in minutes, in the dark, in gloves.
- **It is not a surveillance system.** This deserves plain words, because your community has been asked to adopt camera networks before. This system's radio protocol has a short, fixed vocabulary of fire-defense messages. Video, audio, and personal data have no message type — the network is physically incapable of carrying them. That is architecture, not policy, and the standard's public governance charter forbids changing it. Nothing about your residents leaves their property.

A rugged, palletized version of the same system serves your department directly: stored for years in a warehouse, dropped alongside the pump-and-hose structure protection forestry crews already deploy, it watches the pump site in real time, stretches a fuel load across the whole ember window instead of running dry, and reports back — so nobody drives two hours to find the pump quit six hours ago. Every field unit carries an emergency button that transmits distress by four independent paths, and a charging point, because a firefighter's dead phone at 2 AM is a safety problem, not a convenience problem.

What it changes for your department

A few dozen participating homes, a small drone overhead, and a crew coordinating from a truck at a safe anchor point can hold a neighborhood: the boxes are the hands, the drone is the eyes, and your people stay out of the ember cast. That same drone, flown with a thermal camera, shows you *whether it is working* without anyone leaving the truck: a wetted roof reads cold and a dry one reads hot, so a pass over the neighborhood is a live map of which homes are protected and which have a problem — the coordination picture confirmed against the actual heat coming off the actual roofs. The deployable form of this has a shape your department already understands — a **coordinated defense module**: one Type 6, a trailer of cased units, three or four people, and a drone operator, seeding persistent protection at a threatened community in about an hour and then moving to the next, because the boxes hold the ground after the crew leaves. Nothing about it is new equipment doctrine — it is a new resource type built from resources you already have. And where homes in the community already run their own units, the arriving module's radios join their network automatically: **a prepared community is a cheaper community to defend, measurably.** No system prevents every tragedy, and this one claims no such thing. What it changes is the arithmetic — how many burning places require a person standing there. The West is drying, fire seasons are lengthening, and there will never be enough crews. Ground that defends itself is the honest answer, and every home that holds is a home your people don't stand in front of.

Why open matters to your budget

Commercial coordinated-defense systems are entering the market at $20,000–$80,000 per home with subscriptions, sold through insurance partnerships, priced for your wealthiest residents and no one else. The engineering inside them is standard industrial equipment. An open standard means: your residents build or contract protection at parts-plus-labor cost; your local electricians and irrigation contractors do the work and keep the money in town; every replacement part comes from a normal supply house forever; and no company's bankruptcy, acquisition, or pricing decision can ever strand your community's investment. The standard's charter makes single-source dependency non-compliant by definition. A unit built in 2026 and one built in 2046 will work together.

What we are asking your community to consider

1. **Recognize the standard in your permitting.** Installations are ordinary plumbing and electrical work under existing codes — backflow prevention, NEC compliance, standard permits. A council or district resolution recognizing the specification as an accepted basis for ember-defense installations removes ambiguity for your building department and your residents.
2. **Pilot it.** One street, one water district conversation, one season. The specification includes everything your public works department needs to evaluate coordinated draw against your supply — and the coordination exists precisely to protect your system, not stress it.
3. **Put the field version in your continuity planning.** Prepositioned units are a line item, not a program: no subscription, no activation service, commodity batteries on a shelf-inspection cycle, deployable by whoever is available. And verifying a whole neighborhood of installed units before a season is one overflight, not one property visit at a time — a thermal pass during a test cycle turns the hot roofs into the work order, so maintenance scales to the community instead of the calendar.

4. **Connect it to your existing programs.** Firewise USA communities, community wildfire protection plans, and mitigation grant applications all have natural places for neighborhood-scale coordinated defense. The specification's verifiable event logs are the documentation those programs and your residents' insurers increasingly want.
5. **Send your firefighters' knowledge back.** The standard is governed publicly — anyone may propose improvements, and the people who fight fire are exactly who it needs to hear from. Your department's experience is worth more to this standard than any purchase order.

On the question of proof

Someone at your table will ask — correctly, it's their job — where the studies and certifications are. The direct answer: this is a safety primitive, not a product, and the seatbelt and the smoke detector already ran this experiment. Both were held to proof-before-existence, and the studies that eventually arrived mostly measured the cost of the waiting. Two things make the cautious path even safer here: the system's failure mode is the status quo (a unit that faults degrades to exactly the unprotected house you have today — the baseline is zero, so a pilot risks nothing your community isn't already living with), and the proof instrument ships inside every unit — the verifiable event log makes each deployment a public experiment your department can read. The rigorous move is not to wait for someone else's study. It is to run a pilot street and generate your own.

What this costs your community

The specification, all documentation, and the governance process: nothing, permanently, by design. A resident's installation: roughly the cost of a good appliance. A pilot street: the cost of the units plus your staff's attention. The alternative — waiting for the proprietary version to entrench — costs your community the difference between $1,500 and $40,000 per protected home, forever, and protects only the homes that can pay it.

Who is behind this

The Open Source Firefighter was developed in Colorado by a former firefighter and fire officer with a background in military communications, range operations, and industrial fabrication, and is maintained as a public standard under an open governance charter. It is dedicated to three Colorado firefighters killed on the Knowles Fire during the development of this work. It is free because the people it is built to protect — and the crews it is built to relieve — are not a market. They are the point.

Specification family: main specification, Hardware Annex A, governance charter, and public overview — published under permissive open license.